

CLASS 10 · CBSE · NCERT MATHEMATICS · PYQ SOLUTIONS

Chapter 8: Introduction to Trigonometry

Step-by-step solutions to the previous year questions

The question numbers match the PYQ question sheet.

1 Ratios of standard angles

1. Find the acute angle θ for which $7 + 4 \sin \theta = 9$. **CBSE 2025, 1 mark**

SOLUTION

$$4 \sin \theta = 2, \text{ so } \sin \theta = \frac{1}{2}$$

Answer: $\theta = 30^\circ$

2. Given that $x \cos 60^\circ + y \cos 0^\circ + \sin 30^\circ - \cot 45^\circ = 5$, find the value of $x + 2y$. **CBSE 2025, 2 marks**

SOLUTION

Put in the values: $x \times \frac{1}{2} + y \times 1 + \frac{1}{2} - 1 = 5$

$$\frac{x}{2} + y = \frac{11}{2}. \text{ Multiply both sides by 2.}$$

Answer: $x + 2y = 11$

3. Evaluate $\frac{\tan^2 60^\circ}{\sin^2 60^\circ + \cos^2 30^\circ}$. **CBSE 2025, 2 marks**

SOLUTION

$$\tan 60^\circ = \sqrt{3}, \sin 60^\circ = \frac{\sqrt{3}}{2} \text{ and } \cos 30^\circ = \frac{\sqrt{3}}{2}$$

$$\text{Value} = \frac{3}{\frac{3}{4} + \frac{3}{4}} = \frac{3}{\frac{3}{2}}$$

Answer: 2

4. Given $\cos(\alpha + \beta) = 0$, find $\cos\left(\frac{\alpha + \beta}{2}\right)$. (a) $\frac{1}{\sqrt{2}}$ (b) $\frac{1}{2}$ (c) 0 (d) $\sqrt{2}$ **CBSE 2024, 1 mark**

SOLUTION

$$\cos(\alpha + \beta) = 0 = \cos 90^\circ, \text{ so } \alpha + \beta = 90^\circ.$$

$$\cos\left(\frac{\alpha + \beta}{2}\right) = \cos 45^\circ = \frac{1}{\sqrt{2}}$$

Answer: (a)

5. Find the value of $2\sqrt{2} \cos 45^\circ \sin 30^\circ + 2\sqrt{3} \cos 30^\circ$. **CBSE 2024, 2 marks**

SOLUTION

$$= 2\sqrt{2} \times \frac{1}{\sqrt{2}} \times \frac{1}{2} + 2\sqrt{3} \times \frac{\sqrt{3}}{2}$$

$$= 1 + 3$$

Answer: 4

6. For $A = 60^\circ$ and $B = 30^\circ$, verify that $\sin(A + B) = \sin A \cos B + \cos A \sin B$. **CBSE 2024, 2 marks**

SOLUTION

$$\text{LHS} = \sin(60^\circ + 30^\circ) = \sin 90^\circ = 1$$

$$\text{RHS} = \sin 60^\circ \cos 30^\circ + \cos 60^\circ \sin 30^\circ = \frac{\sqrt{3}}{2} \times \frac{\sqrt{3}}{2} + \frac{1}{2} \times \frac{1}{2} = \frac{3}{4} + \frac{1}{4} = 1$$

LHS = RHS. Verified.

2 Trigonometric identities

7. Simplify $\tan^2 \theta - \left(\frac{1}{\cos \theta} \times \sec \theta \right)$. **CBSE 2025, 1 mark**

SOLUTION

$$\frac{1}{\cos \theta} = \sec \theta, \text{ so the bracket is } \sec^2 \theta.$$

$$\tan^2 \theta - \sec^2 \theta = -(\sec^2 \theta - \tan^2 \theta) = -1$$

Answer: -1

8. Prove that $\frac{\tan \theta}{1 - \cot \theta} + \frac{\cot \theta}{1 - \tan \theta} = 1 + \sec \theta \operatorname{cosec} \theta$. **CBSE 2025, 3 marks**

SOLUTION

Write everything in terms of $\sin \theta$ and $\cos \theta$.

$$\text{First term: } \frac{\sin \theta}{\cos \theta} \div \frac{\sin \theta - \cos \theta}{\sin \theta} = \frac{\sin^2 \theta}{\cos \theta (\sin \theta - \cos \theta)}$$

$$\text{Second term: } \frac{\cos \theta}{\sin \theta} \div \frac{\cos \theta - \sin \theta}{\cos \theta} = -\frac{\cos^2 \theta}{\sin \theta (\sin \theta - \cos \theta)}$$

$$\text{LHS} = \frac{\sin^3 \theta - \cos^3 \theta}{\sin \theta \cos \theta (\sin \theta - \cos \theta)}$$

$$\text{Using } a^3 - b^3 = (a - b)(a^2 + ab + b^2): \text{LHS} = \frac{\sin^2 \theta + \sin \theta \cos \theta + \cos^2 \theta}{\sin \theta \cos \theta} = \frac{1 + \sin \theta \cos \theta}{\sin \theta \cos \theta}$$

$$= \frac{1}{\sin \theta \cos \theta} + 1 = 1 + \sec \theta \operatorname{cosec} \theta = \text{RHS. Proved.}$$

9. Prove that $\frac{\sin A + \cos A}{\sin A - \cos A} + \frac{\sin A - \cos A}{\sin A + \cos A} = \frac{2}{2\sin^2 A - 1}$. **CBSE 2025, 3 marks**

SOLUTION

Take the common denominator $(\sin A - \cos A)(\sin A + \cos A) = \sin^2 A - \cos^2 A$.

Numerator = $(\sin A + \cos A)^2 + (\sin A - \cos A)^2 = 2\sin^2 A + 2\cos^2 A = 2$

$$\begin{aligned} \text{LHS} &= \frac{2}{\sin^2 A - \cos^2 A} = \frac{2}{\sin^2 A - (1 - \sin^2 A)} \\ &= \frac{2}{2\sin^2 A - 1} = \text{RHS. Proved.} \end{aligned}$$

10. Given $\sec \theta - \tan \theta = m$, find $\sec \theta + \tan \theta$. (a) $1 - \frac{1}{m}$ (b) $m^2 - 1$ (c) $\frac{1}{m}$ (d) $-m$ **CBSE 2024, 1 mark**

SOLUTION

$\sec^2 \theta - \tan^2 \theta = 1$, so $(\sec \theta + \tan \theta)(\sec \theta - \tan \theta) = 1$.

$$\sec \theta + \tan \theta = \frac{1}{\sec \theta - \tan \theta} = \frac{1}{m}$$

Answer: (c)

11. Prove the identity $\frac{\tan \theta}{1 - \cot \theta} + \frac{\cot \theta}{1 - \tan \theta} = 1 + \sec \theta \operatorname{cosec} \theta$. **CBSE 2024, 3 marks**

SOLUTION

Write $s = \sin \theta$ and $c = \cos \theta$. Then $\frac{\tan \theta}{1 - \cot \theta} = \frac{s/c}{(s-c)/s} = \frac{s^2}{c(s-c)}$ and $\frac{\cot \theta}{1 - \tan \theta} = \frac{c/s}{(c-s)/c} = \frac{c^2}{s(c-s)}$.

$$\begin{aligned} \text{LHS} &= \frac{s^3 - c^3}{sc(s-c)} = \frac{(s-c)(s^2 + sc + c^2)}{sc(s-c)} = \frac{s^2 + c^2 + sc}{sc} \\ &= \frac{1 + sc}{sc} = \frac{1}{sc} + 1 \\ &= 1 + \sec \theta \operatorname{cosec} \theta = \text{RHS} \end{aligned}$$

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